

AMRIT SINGH

AI/ML Engineer & Full-Stack Developer

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PROFESSIONAL SUMMARY

AI/ML Engineer & Full-Stack Developer with expertise in **Python**, **Machine Learning**, **React.js** and **FastAPI**. Passionate about building scalable web applications and AI-powered solutions with clean, efficient code.

WORK EXPERIENCES

Research Intern | Internship

Nov 2025 - Apr 2026

Samsung R&D

- Developed an AI-powered platform that generated visual appliance repair guides from user-reported faults and technical specifications. Reducing manual troubleshooting time by **50%**.
- Built a RAG pipeline using **LangChain**, **ChromaDB**, and **HuggingFace** to retrieve instructions and diagrams from PDF manuals.
- Engineered a React web app with optimized vector search and automated document parsing using Vision LLMs.

Technologies / Skills Used : Python, Flask, React, Express, LLMs, RAG

Web Developer Intern | Internship

Sep 2025 - Nov 2025

Techsolid Development

- Built responsive web pages and reusable UI components using **HTML**, **CSS** and **JavaScript**.
- Developed backend logic and **RESTful APIs** in **Python** to support core application features.
- Collaborated across the full stack to seamlessly integrate frontend interfaces with **FastAPI** microservices

Technologies / Skills Used : Python, HTML, CSS, JavaScript, FastAPI

EDUCATIONS

International Institute of Information Technology Naya Raipur (IIITNR) | Chhattisgarh

Aug 2023 - May 2027

B.Tech in Electronics & Communication Engineering

CGPA: 8.75/10

PROJECTS

AI-Powered Manim Video Generator | 🐙 Github 🌐 Website

May 2026 - Present

- Created a platform that converts text prompts into animated educational videos using **AI-generated Manim code**.
- Implemented **LLM-powered prompt refinement**, script generation, code validation, asynchronous rendering, and cloud-based video storage.

Technologies / Tools Used : FastAPI, Celery, Redis, PostgreSQL, Cloudinary, Vercel, LLM, GCP

ModuLearn - AI-Driven Adaptive Learning Platform | 🐙 Github

Jan 2026 - Apr 2026

- Developed an **AI-powered learning platform** generating personalized study materials from topics, PDFs, and web content.
- Implemented adaptive learning workflows and interactive features to boost user engagement.
- Developed AI-powered educational content workflows, including video summaries, quizzes, flashcards, and personalized YouTube recommendations.

Technologies / Tools Used : React, LLM, Supabase, Data Preprocessing, FastAPI

Text-to-SQL table | 🐙 Github

May 2025 - Dec 2025

- Developed a **hybrid Text-to-SQL table name retrieval pipeline** combining **BM25** with **BAAI/bge-base-en-v1.5** semantic embeddings indexed using **FAISS** for accurate schema retrieval.
- Implemented **Cross-Encoder reranking** using **ms-marco-MiniLM-L6-v2** to re-score top candidates with joint query-table attention, improving retrieval precision to **92%+ Top-5 accuracy**.

Technologies / Tools Used : FAISS, BM25, BAAI/bge-base-en-v1.5, Sentence-Transformers, Cross-Encoder, scikit-learn.

SKILLS

Programming Languages :

Python, JavaScript, C++

AI/ML :

Pandas, NumPy, LangChain, Hugging Face, TensorFlow, PyTorch, LLMs, EDA

Data Visualization: :

Matplotlib, Seaborn, Plotly

Frameworks & Databases :

React, FastAPI, Celery, PostgreSQL, MySQL, Redis

Tools & Platforms :

Git, Docker, Postman, CI/CD